

Why do Banks exist?

Introduction

Banking appears to have significant trade-offs. On the one hand, it is a source of great economic value, with banks channelling savings into investment portfolios and being one of the most significant sources of wealth creation (King and Levine 1993). However, they are also prone to collapsing to the detriment of the economy and often require government assistance to stay afloat. Both the papers I will analyse explain the two cornerstones of banking: liabilities with Diamond (1984) and assets with Diamond-Dybvig (1983), and both assess the fundamental question of why banks exist.

Diamond (1984) addresses this in his paper, explaining that their existence depends on their ability to solve the problem of ‘free riders’ by having banks act as delegated monitors. I will assess the insights it provides and explain its impact on the financial world. Similarly, I will do the same for Diamond and Dybvig (1983). The paper focuses on why banks are so vulnerable to collapse and how they exist, emphasising liquidity mismatches. Diamond and Dybvig (1983) explain this via the issue of bank runs.

The scope of this essay will relate my empirical assessment of the contributions of both papers to the 2008 financial crisis and show how the papers both explained aspects of the crisis and how if the suggestions from the models are incorporated into the financial system how this can create a better or worse financial system and one less or more prone to collapses in order to critically assess the contribution of the papers in its practical implications. In doing so, I will initially explain the paper's theoretical contributions and its impact on the financial literature to understand its practical outcomes fully.

Diamond (1984)

Diamond (1984), in his paper “Financial Intermediation and Delegated Monitoring”, focuses on the issue of ‘free riders’ in the financial system. It occurs when the ‘free rider’, a lender, benefits from monitoring a borrower but does not pay for the cost of that monitoring. His solution to the issue is seen as a key contribution of the paper and explains why banks exist.

Key Terminology

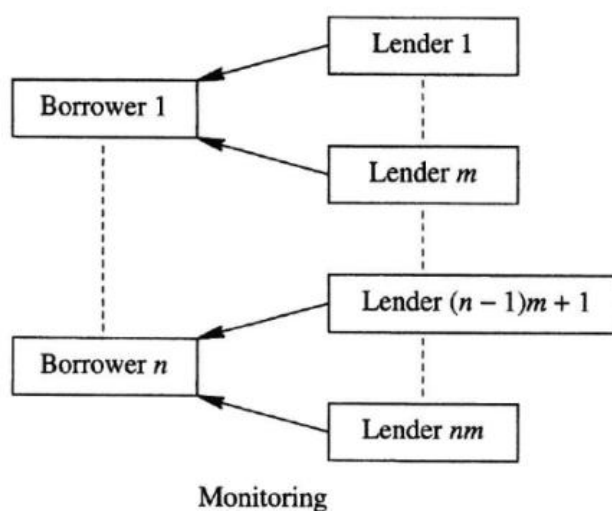
Monitoring entails observing a borrower’s actions and returns in accordance with the terms of the loan contract. The entrepreneur is an individual with an idea with insufficient capital, the lender is a risk-neutral investor providing loans, the intermediary is a financial agent (bank) who is delegated to monitor these loan contracts, and financial intermediation is the process of connecting surplus fund holders to those who need funds (Diamond 1984).

Why does monitoring need to happen, and who benefits?

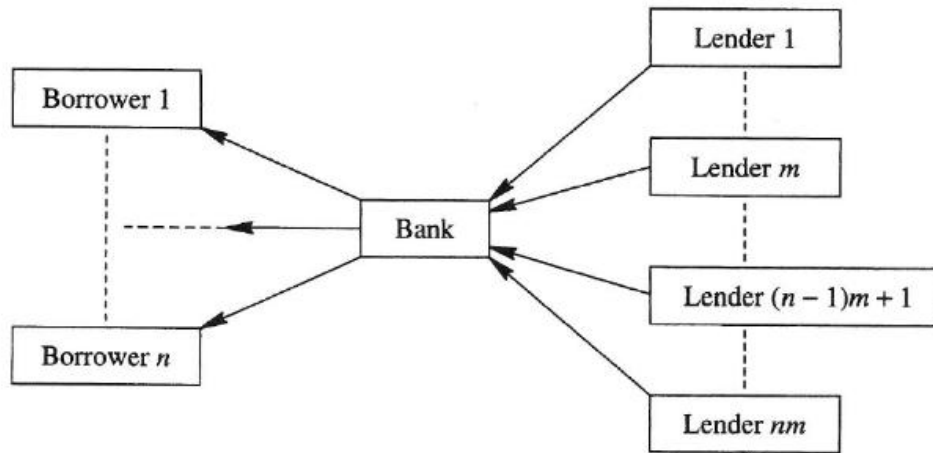
Lenders are risk-neutral and rational, with wealth of $1/m$, where m is the number of entrepreneurs, and will lend their money if the expected return is greater than or equal to R , which, for example, could be the rate on a national savings account. The financial intermediary is risk-neutral, has no wealth, and incurs a monitoring cost of K for all entrepreneurs, with N depositors (Diamond 1984). The reason the entrepreneur does not lie about how much money the project has generated, saying it generated nothing, is that they will receive nonpecuniary penalties for doing so, such as reputation loss, which can lead to restricted access to capital in the future. This leads the entrepreneur not to have an incentive to report that they made nothing from the project, thereby giving lenders greater confidence to invest (Diamond 1984). Without the intermediary, the entrepreneur tries to maximise their personal return, net of nonpecuniary penalties. Since lenders get a rate greater than or equal to R , they are still willing to lend. However, with the intermediary, the intermediary monitors all lenders, once only spending K to provide information on the project's actual outcome. They collect funds from many lenders and provide an expected return greater than or equal to R , lower than the rate they loan, with the difference serving as a monitoring fee, and their return being greater than the cost of monitoring. It means that the more clients the bank has, the more the cost of monitoring is spread (Diamond 1984).

Figures 1.1 and 1.2 (Brunnermeir)

Direct lending with monitoring/2



Delegated monitoring/2



As shown in Figures 1.1 and 1.2, without the bank, lenders individually monitor each borrower; however, when the bank is introduced as a delegated monitor, it splits the fixed cost exactly across borrowers.

Who Monitors the Monitor?

There is no need for a monitor of the monitor as it becomes unprofitable and unnecessary for the bank to lie as total loans become more predictable as the number increases due to diversification wherein the bank lends to many independent borrowers to spread the risk of individual loans making their income more stable and reducing the need for costly monitoring or penalties as the cost is spread (Diamond 1984). It means the bank can safely promise a fixed payment, such as deposit interest, to depositors. Since the probability of total shortfall is small, the bank rarely faces defaults or penalties. This leads to the bank having no incentive to misreport or shirk monitoring, as it risks losing its reputation, crucial as it would mean fewer depositors would trust the bank for little or no gain, with the bank being more efficient with more depositors, as the model displays a perfect monopolistic model, wherein the most efficient outcome involves a monopoly (Diamond 1984).

Contribution to the literature

One aspect that Diamond (1984) had been credited shifting the academic study of banking as banks which were previously seen as primarily functioning to move money and were now primarily seen as

information producers for monitoring within Information Economics. It led to decades of research into banks as solutions to information asymmetries of moral hazard, which is the lack of incentive to guard against risk when one is protected from it and adverse selection, which is when, during asymmetric information, one party takes advantage of undisclosed information to gain an advantage (Freixas and Rochet 2008). He also explained diversification in a new way, emphasising that banks diversify to create a more predictable portfolio, so depositors do not need to monitor the bank, leading to trust between the parties. It can be argued that the market is better and more efficient at solving the information problem than banks. This is called 'disintermediation' when borrowers bypass banks and go straight to the market (Boot and Thakor 1997). With the rise of the corporate bond market, contrary to Diamond (1984), who suggested borrowing should be facilitated via banks. Many large firms began issuing these corporate bonds, such as General Electric. This happened with the rise of credit rating agencies, which acted as monitors. It is argued that a credit score or rating is a better form of monitoring, and large firms do not need loans; they can just issue bonds. Then the credit rating agencies assess the risk, and people can lend to large firms, crucially, only available to large firms due to information costs, split similarly to figures 1.1 and 1.2, and reputation, similarly to banks. Diamond himself affirmed this change and adjusted his research to encompass the new system, creating a model of disintermediation (Diamond 1991).

Practical Implications

The 2008 financial crisis underscored the dangers of failing to monitor lenders properly. It also showed that systems without monitors could exist, even though they are less Pareto efficient. This occurred when a shadow banking system was created by mortgage brokers who would create subprime loans to lenders, which would then be sold to investment banks, with the investment banks bundling them into mortgage-backed securities, selling these to investors, with Credit Rating Agencies acting as delegated monitors who would rate investment banks for how credible their investments were (Gorton, 2009). The issue was that brokers also had no incentive to give credible loans as they were incentivised to maximise loan volume to earn commission (Keys et al. 2010). As well as the monitor (Credit Rating Agencies) having no incentive to pay the price for monitoring because their money was not at risk if their ratings proved to be wrong a form of moral hazard, particularly as the investment banks paid them and not the investors which created an incentive to give favourable ratings to win more business rather than being credible monitors for the investors (Bolton et al. 2012). This led to a modification of the free rider problem in Diamond's model. The investor, for example, a pension fund investor, had no way to monitor the credibility of the loans they were invested in, such as European investors having no way to monitor mortgages in the United States. When house prices stopped rising and began to fall, interest rates also began to increase, which

led to mortgage payments increasing and because of the lack of screening of mortgage borrowers many could not afford to pay for these increases and had led to a substantial portion of people defaulting on their mortgages and mortgage back securities failing being a significant trigger for the 2008 financial crisis. Diamond (1984) had exposed the crucial need for credible delegated monitors within the financial system due to their being ignored due to moral hazard, the crucial role monitors play and the severe outcomes that could occur if monitoring was ignored, particularly in systems less regulated than banking, 24 years after his paper had been published. An example of how his model could have been incorporated is that, if brokers had to retain a portion of the loans, they would have been incentivised to monitor their customers. Whilst the failure of monitoring created toxic assets, it was the structure of bank liabilities that converted these asset losses into systemic panic, which is a phenomenon explained well by Diamond and Dybvig (1983). Diamond (1984) shows that even without analysing liabilities, banks may still exist just to solve the information problem.

Diamond and Dybvig (1983)

This model explains how banks can be vulnerable to bank runs, a phenomenon in which fearful bank depositors withdraw their money from the bank as they fear the bank collapsing and not getting their money back. It flips the way they are understood from irrational panic to a rational self-fulfilling prophecy. This, in turn, leads to the bank being more vulnerable to collapsing due to liquidity mismatch, which is when an entity's assets cannot be converted into cash to meet short-term liabilities. A key contribution of the paper was that it showed that what made them so fragile, transforming illiquid assets into liquid deposits, was also what made them essential for the economy.

Diamond Dybvig have a model of an economy where individuals have uncertain liquidity needs. Some people need their money earlier and are impatient, and others can wait and are patient. A bank serves the purpose of pooling both individuals' savings, investing their money in long-term illiquid projects such as mortgages or business loans, which offer high returns but cannot be exchanged early without significant losses incurred. Meanwhile, the bank offers its depositors the ability to withdraw their money on demand. This maturity transformation is important as it allows patient savers to fund productive long-term investments and provides liquidity for impatient savers.

This leads to a coordination game with two Nash equilibria, outcomes where it is not rational to deviate. The good equilibrium is where the depositors believe the bank is stable, where most people leave their money in, and the people with immediate liquidity needs withdraw, with everyone better off. However, the bad equilibrium (the bank run) occurs when depositors fear that other depositors will mass withdraw their money. This fear, whether based upon valid concerns or speculation, makes it individually rational to rush to withdraw money.

Figure 2.1 (Balkenborg et al. 2021)

		Depositor 1	
		Today	Tomorrow
Depositor 2	Today	750 750	0 1000
	Tomorrow	0 1000	1500 1500

$R=1.5, L=.5$

With R being the long-term return, which is the gross return on the bank's assets allowed to mature. Whereas L is the liquidation value or fire-sale value, and is the return in the case of a bank run, where the bank is forced to sell assets early. Figure 2.1 shows how a bank run is rational. If depositor one runs today, while depositor two waits, depositor 1 gets their entire 1000 back, and because the fire-sale price is so low, the bank has to sell everything just to pay the first withdrawer, leaving 0 for the second depositor. This is known as the first-mover advantage. In the bad equilibrium, both depositors run as they do not want to be left with zero, and then the bank liquidates everything, and they end up with 750 in the bad equilibrium (bank run), half the value compared to the 1500 they could have had if they both waited until tomorrow in the good equilibrium. In moving from the bad equilibrium to the good equilibrium, Diamond and Dybvig proposed government-backed deposit insurance. This entails a credible guarantee that insures depositors against losses, ensuring that all depositors will get their money back and removing the incentive for depositors to run.

Contribution to the literature and the financial system

Diamond and Dybvig (1983) changed the way bank runs were viewed; initially, they were seen as episodes of mass hysteria and irrationality on the part of consumers. Diamond and Dybvig formalised

the idea that bank runs were the rational option for consumers, whether their fears about the bank were genuine or not (Allen and Gale 2007). It can be argued further that Diamond-Dybvig's view that bank runs are unfounded and based solely on panic is unrealistic. Gorton (1988) studied historical bank panics in the U.S. and found they were not random events and were highly predictable and almost always occurred at the peak of the business cycle, just about when the economy was about to enter a recession. He argued that depositors ran because they correctly anticipated a recession would cause loans to fail, making the bank insolvent, showing the run as a symptom of a failing economy and banks not the cause of the failure. Calomiris and Gorton (1991) then rectify this view and argue that there is a trigger that makes depositors nervous enough to consider withdrawing from the bank. Once that happens, the Diamond-Dybvig coordination problem determines whether the run is successful. Showing how deposit insurance stops one of the equilibria from being possible, and showing how Diamond-Dybvig advanced the literature on understanding bank runs

Practical Implications

Similar to the Diamond 1984 model, the model gained more credibility after the 2008 financial crisis and exposed the risk of banks running without insurance existing in the shadow banking system. The organisations that acted like banks were Money Market Funds, which were used by millions of individuals like high-yield savings accounts (Gorton and Metrick 2010). They were priced at a stable 1 dollar per share and redeemable at any time, like a bank deposit. To pay for the higher yield via their illiquid assets, they invested in short-term corporate debt that was a risky long-term investment (Kacperczyk and Schnabl 2013). Then, in 2008, the investment bank Lehmann Brothers filed for bankruptcy. The reserve primary fund had 785 million dollars of Lehmann's commercial paper. This led the debt to become worthless and led to shares only being worth 97 cents (Kacperczyk and Schnabl 2013). This then triggered a bank run with investors rushing to withdraw their assets from other funds too, as they feared that they were also invested in toxic debt, and within 2 days, 300 billion dollars was withdrawn from accounts. This caused funds to sell their assets at low prices to meet withdrawal demands (Kacperczyk and Schnabl 2013). This was only halted when, as recommended in Diamond Dybvig, the United States Treasury stepped in to guarantee the 1-dollar valuation (Kacperczyk and Schnabl 2013).

When a credible institution creates deposit insurance, which can incur any potential losses, such as the U.S. Treasury's intervention in the 2008 financial crisis, it can lead to an immediate halt in a bank run, with investors having confidence in the U.S Treasury paying out. This led to the bad equilibrium ending and stopped the effects of financial contagion as financial markets would, over time, stabilise (Allen and Gale 2000). Financial contagion occurs when banks are connected by interbank lending (lending to smooth out small liquidity shocks). It can be defined as the phenomenon where a slight

liquidity shock or preference shock in one region can spread throughout the economy, and can be explained through the fact that in bad times, banks will liquidate claims on each other, forcing the other bank to potentially liquidate assets to pay for the claim (Allen and Gale 2000).

Eventually, the programme prevented an estimated \$3 trillion in withdrawals, which would have had catastrophic consequences (Congressional Research Service 2020). The programme was so successful that the guarantee was never paid out on a large scale. The U.S. Treasury collected fees from funds for this insurance. Eventually, it led to the U.S. Treasury a profit of \$1.2 billion and was widely hailed as a success and contributed to the designation of the paper for the Nobel prize, with it being hailed as preventing a second great depression (Congressional Research Service, 2020)(Committee for the Prize in Economic Sciences in Memory of Alfred Nobel, 2022). Showing how deposit insurance can be beneficial for all parties involved, how it needs credible institutional backing, how institutions can create deposit insurance during a bank run, and how it remains effective in changing the equilibrium.

Practical Implications of Moral Hazard

This deposit insurance can also be detrimental to the economy. As depositors are protected against bank failure, they become less careful when choosing where to deposit their money, and they will choose a protected bank with the highest return (Demirgüç-Kunt and Detragiache 2002). This can lead to moral hazard, as it gives banks a stronger incentive to take on higher risk, which can involve lending and investing, and leads them to monitor their clients less, with catastrophic consequences, as demonstrated by Diamond (1984).

This is shown empirically as it exacerbated the Saving and Loan crisis in the 1980s wherein riskier investments by saving and loan institutions were prompted by deposit insurance, many of these investments had then failed and then the Federal Saving and Loan Insurance Corporation had gone bankrupt trying to recover the losses, which led to the failure of many savings and Loan companies (Kane 1989). This shows that without credible institutions capable of creating a sustainable deposit insurance model, it will fail and become a negative. However, if institutions are capable and willing, whatever effect of moral hazard that exists, for example, risky loans and investments, is offset by the positive effect of averting very destructive bank runs (Demirgüç-Kunt and Detragiache 2002).

Although it can be argued that bank runs can be a positive and play a disciplinary role in the institution, and that they are the only thing that keeps banks honest. This view is advanced by Calomiris and Kahn (1991), who argue that the first-come, first-served provision in a bank contract is a feature, not a flaw. If depositors see bankers engaging in moral hazard, such as making risky loans, they can run on the bank and get their money before it fails. The threat of this happening, they argue, would stop bankers from engaging in moral hazard. Therefore, deposit insurance would be a negative.

However, Rochet and Vives (2004) argue that a disciplined run by a few informed investors can lead to financial contagion, as the uninformed public may misinterpret it as a Diamond-Dybvig-style panic. This is exemplified by the 2008 financial crisis as the collapse of Lehmann Brother's had triggered a bank run on Money Market Funds who acted like banks regardless of whether they had unsafe assets or not and shows the effect of financial contagion as even if as Gorton (1988) had argued that bank runs only happen in economic bubbles but this does not negate that even well invested banks are safe from the effects of financial contagion triggered by bank runs and shows an effect similar to the findings of Calomiris and Gorton (1991). Meaning that banks would still engage in moral hazard if they feared financial contagion as if other banks start taking risky bets, then there is no incentive not to take risky bets as it would put a bank at a disadvantage, as they would face similar repercussions for taking risky bets compared to not taking them and would have lower returns if they did not take them. It is argued that the moral hazard from fear of financial contagion outweighs that from deposit insurance and entails the risk of devastating bank runs (Acharya 2009).

Conclusion

In conclusion, the contributions of both papers have been highly significant within the economic landscape and have advanced policy and research by explaining various aspects of the banking sector. Diamond (1984) highlighted the role of banks as delegated financial monitors; he advanced the literature in this regard, as his contribution would lead to decades of research on banks as information gatherers assessing their moral hazard and adverse selection. It was also shown clearly during the 2008 financial crisis wherein third parties who had no incentive to monitor were given the role in a shadow banking system with the consequences of not monitoring due to moral hazard being catastrophic as it would lead to many people defaulting on their mortgages when interest rates increased as they were not appropriately screened and led to many mortgage backed securities to fail triggering the 2008 financial crisis. The Nobel Prize-winning Diamond-Dybvig (1983) showed that, in the literature, bank runs were rational for bank customers to participate in. The reason they exist is to address liquidity mismatches; when this fails due to bank runs, it underscores the need for banks to be backed by deposit insurance. This was shown by the devastating consequences avoided in the 2008 financial crisis due to the U.S. government implementing deposit insurance. However, with the drawback that deposit insurance can cause moral hazard, which leads to riskier loans and investments from banks, despite this drawback, it is argued that the avoidance of bank runs in this model is more Pareto efficient and may also be less than the moral hazard without deposit insurance due to the presence of financial contagion. These factors show that, despite some drawbacks, the benefits of each model outweigh its costs, and they both have significantly contributed to the financial literature.

Particularly when in conjunction to explain the two cornerstones of banking assets and liabilities and have been proven true by empirical examples in the financial landscape, years after their publication.

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